Aidan Howard

Design, Build + Research Projects

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Projects at a Glance

Six projects, ordered as they appear in this portfolio

01 DIMLab Hip Exoskeleton

Univ. of Dayton

Defined the motor misalignment problem and developed active alignment for a rehab exoskeleton

Skills: MATLAB simulation, SolidWorks, research

02 Multifunctional Wrench Handle

UMass Amherst

Handle for interchangeable wrench heads, checked with hand calcs and FEA

Skills: Strength of materials, ANSYS FEA, CAD

03 Modular Robotic Fish

RIT Hydrodynamic Lab

Waterproof Arduino-controlled fish for testing seal-whisker flow sensors

Skills: Iterative prototyping, 3D printing, soldering

04 Miniature Piston

RIT

Small-scale piston assembly drafted and machined from my own drawings

Skills: Mill, lathe, engineering drawings

05 Freestanding Climbing Wall

Personal Project

Climbing wall designed to stand without bolting into the walls of my home

Skills: Power tools, basic structural design

06 Automated Chime Machine

RIT

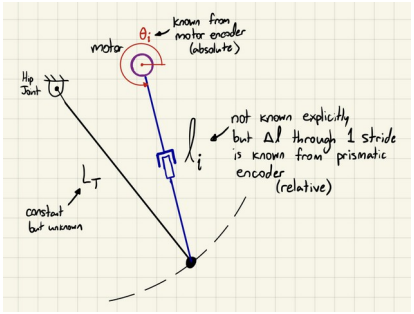
Led a 4-person team to design, fabricate, and program an automated instrument

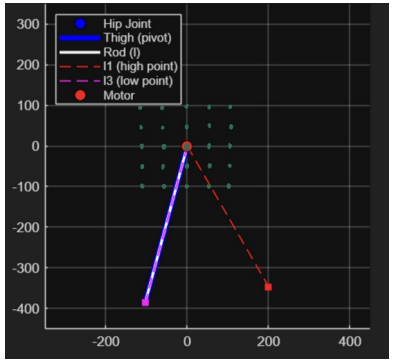
Skills: Team leadership, fabrication, mechanisms

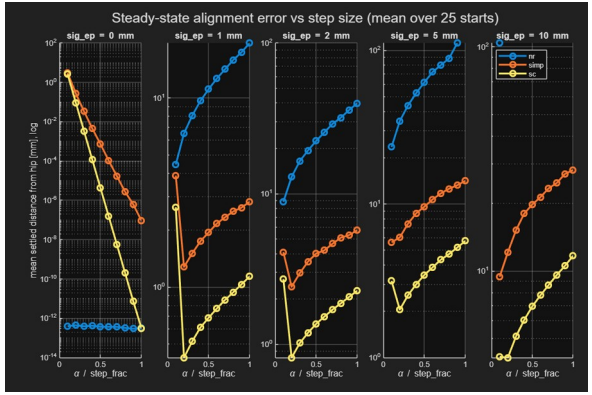
DIMLab Hip Exoskeleton

Univ. of Dayton, Summer 2026 | Defined the misalignment problem and developed active motor-to-hip-joint alignment

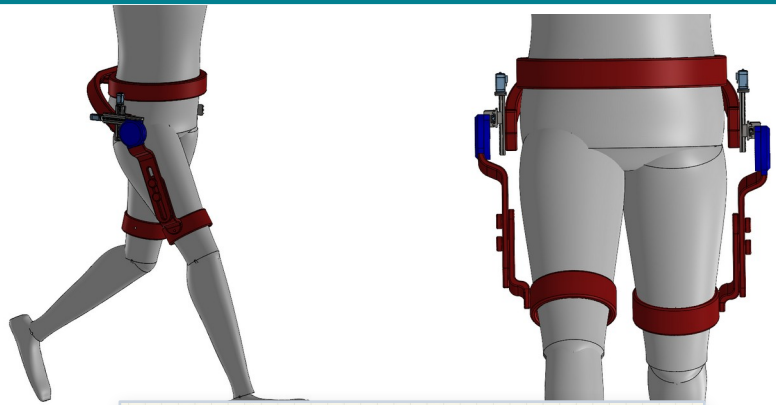
Algorithms + Data Analysis + MATLAB Simulation

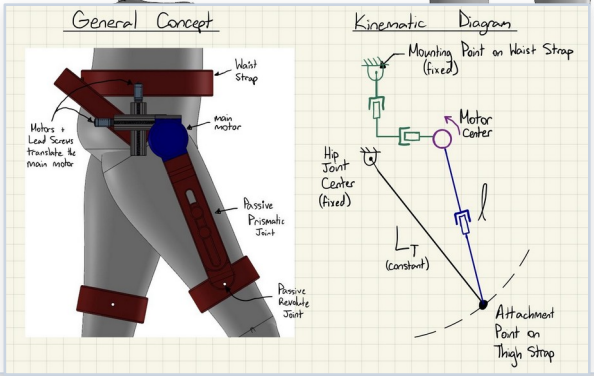






CAD Modeling

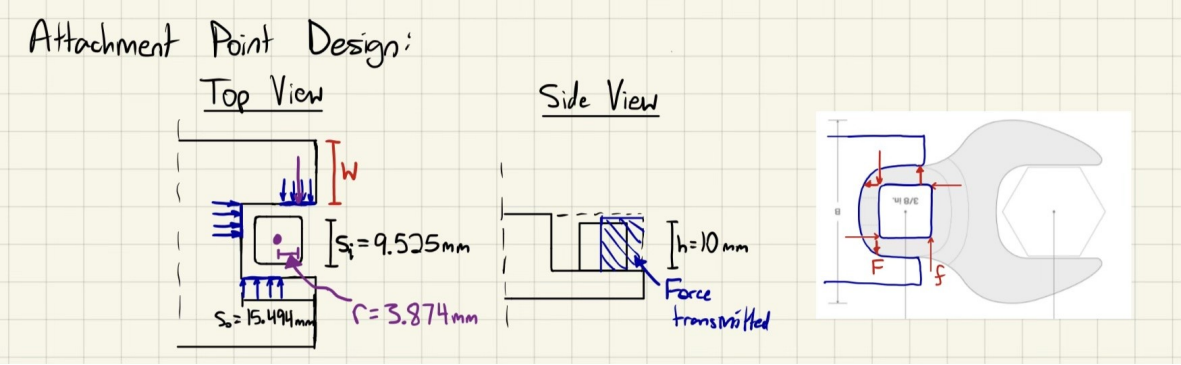




Multifunctional Wrench Handle

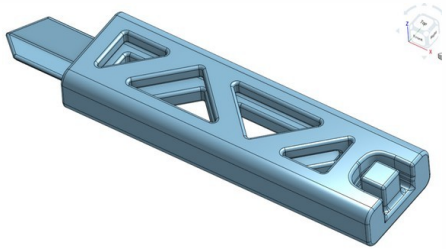
UMass Amherst | Load case derived from published torque-tension data; ANSYS FEA validated against physical testing

Load Modeling

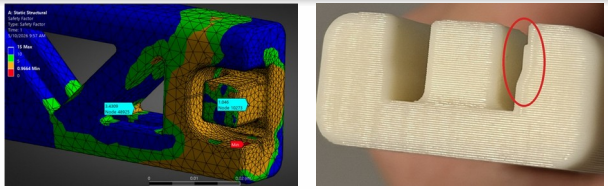


Modeled the applied torque as opposing face loads on the socket walls over the 10 mm of contact depth, setting the FEA load case.

CAD Model



ANSYS FEA vs. Physical Test



Minimum safety factor of 0.97 at the head pocket matched the slight plastic deformation after testing (circled)

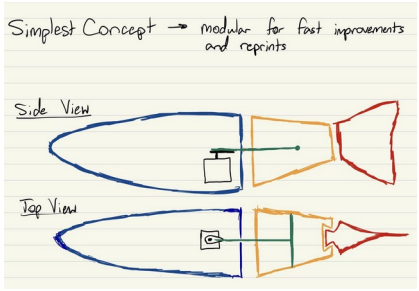
Physical Test



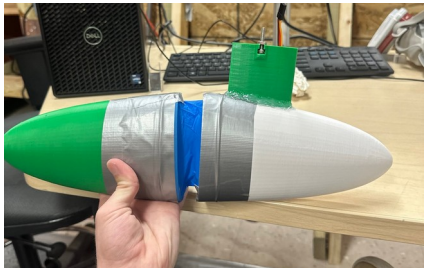
Modular Robotic Fish

RIT Hydrodynamic Lab | Led design + fabrication of an Arduino-driven fish that simulates wakes for sensor testing

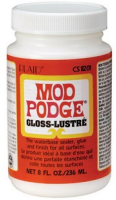
Concept



Fabrication

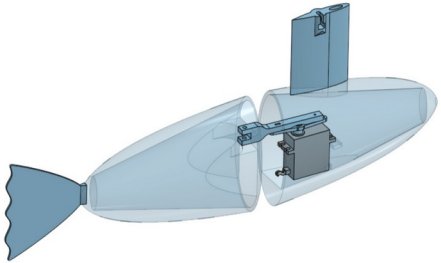


Revision

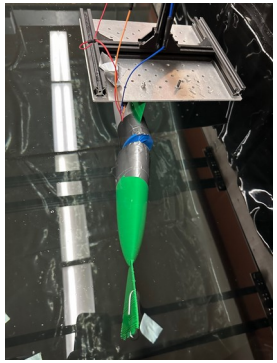


(Water seeped through layers in the plastic during testing; sealed all printed parts with Mod Podge)

CAD Model



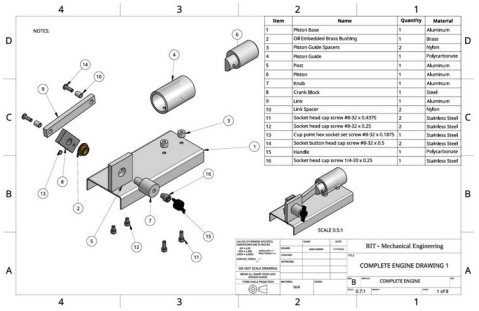
Testing

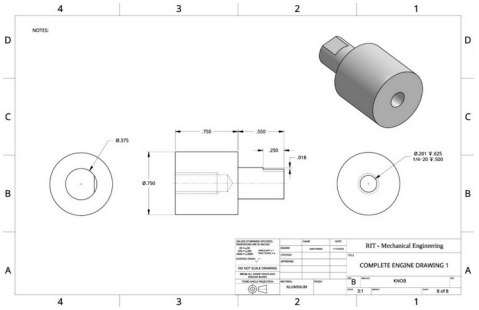


Hands-On Builds: Miniature Piston + Climbing Wall

RIT: drafted piston drawings, then machined parts on mill + lathe | Personal: freestanding wall built with power tools

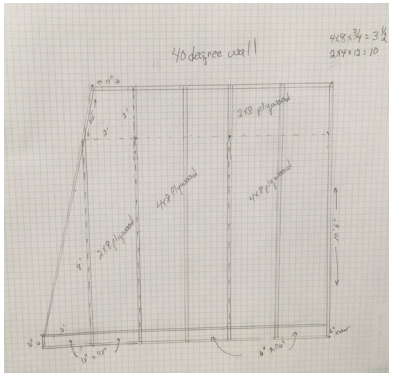
Miniature Piston Project







Freestanding Climbing Wall



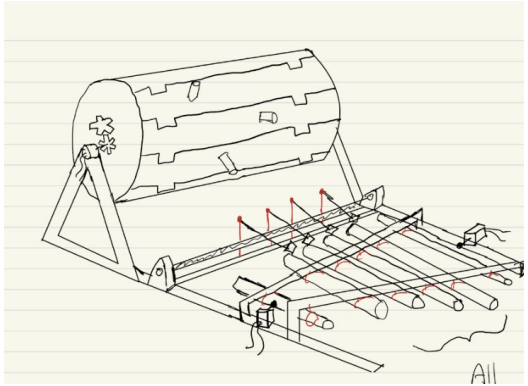




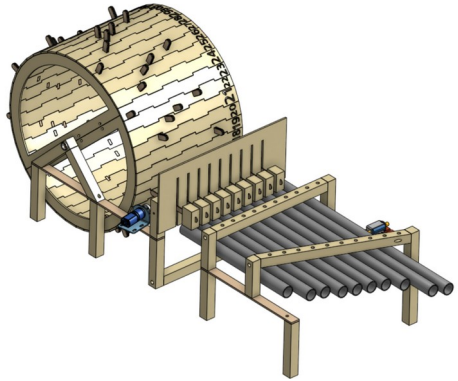
Automated Chime Machine

RIT | Led a 4-person team to design, fabricate, and program an automated musical instrument

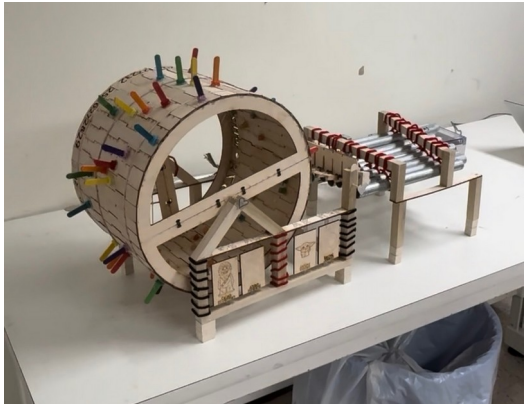
Concept



CAD Model



Fabrication



Outcome: Selected by professors and TAs as a showcase project for the ImagineRIT innovation exhibition

Thank You

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